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Chapter 1

Mobile GIS overview and mapcentric data collection using ArcGIS Field Maps

Objectives

- Grasp the Mobile GIS concept and advantages.
- Understand outdoor and indoor positioning technologies.
- Understand the four key pillars of Mobile GIS capabilities.
- Use Field Maps Designer to create feature layers and layer schemas.
- Design smart forms using Arcade.
- Create feature templates to streamline data entry.
- Collect data using the Field Maps mobile app.
- Review data using Map Viewer.

Introduction

In the post-PC era, mobile devices have become integral to both our personal lives and our professional workflows. Many industries have adopted the "mobile first" strategy, recognizing mobile platforms as the primary interface for enterprise information systems. This is especially true for geographic information systems (GIS), in which the location-aware capabilities and other advantages of mobile platforms are particularly valuable. Mobile GIS has revolutionized the way we acquire, visualize, analyze, and disseminate geospatial information, making it an essential component of contemporary GIS architecture and applications.

This chapter delves into the concepts and advantages of Mobile GIS, including mobile positioning technologies and its architecture within modern GIS. It outlines the four main pillars, or capabilities, of Mobile GIS: planning and workforce coordination, data capture, field awareness, and integration with enterprise systems. Furthermore, the chapter introduces the suite of ArcGIS® responsive web apps and native mobile apps, highlighting the Field Maps component and its basic workflow.

The tutorial section guides you through the role of a solution creator, teaching you how to create hosted feature layers, design layer schemas, build smart forms, style layers, and create feature templates using Field Maps Designer. Additionally, the tutorial positions you in the role of a mobile worker, using the Field Maps mobile app to capture data points, lines, and polygons, and reviewing the collected data through ArcGIS Online Map Viewer and a dashboard.

Mobile GIS: Concepts and advantages

The mobile platform presents a new paradigm for GIS. Mobile GIS, which refers to the use of GIS on mobile devices, originated in the mid-1990s to support field operations, such as surveying and utility maintenance. Building on the widespread availability of mobile devices, mobile networks, and cloud computing, Mobile GIS has become a crucial and widely adopted platform in modern GIS systems.

The advantages of Mobile GIS over traditional desktop GIS are as follows.

- **Mobility:** Mobile devices are wireless, allowing the extension of GIS to areas where traditional wiring is impractical or expensive. This enhances situational awareness in the field and facilitates the transfer of GIS data back to the office.
- **Location awareness:** Technologies, such as GPS, cellular networks, Wi-Fi, and Bluetooth, enable precise location tracking of mobile devices. Additional sensors such as compasses, gyroscopes, and motion sensors help determine the device's orientation, tilt, and speed.
- **Ease of data collection:** Mobile GIS eliminates the error-prone paper-based methods traditionally used in field surveys by digitizing data collection, thereby reducing costs and improving data accuracy.
- **Near-real-time information:** Mobile networks provide a live connection that enhances the temporal capabilities of GIS, enabling continuous monitoring of both spatial and temporal changes in the environment.
- **Versatile communication tools:** Mobile devices integrate with various communication methods, including voice calls, text messages, photos, videos, emails, and social networking apps, facilitating effective collaboration and communication among professionals and the public.
- **Widespread accessibility:** The ubiquity of smartphones, tablets, and smartwatches has made GIS accessible to billions, significantly expanding the user base and applications of GIS.

Although this book primarily focuses on the enterprise applications of Mobile GIS, it is important to recognize that Mobile GIS serves a broad spectrum of uses for individual consumers and enterprise organizations. Consumers frequently use Mobile GIS for everyday activities such as locating nearby points of interest, finding restaurants, navigating routes, and sharing experiences with others. Meanwhile, enterprise organizations depend on Mobile GIS for critical functions, such as data visualization, field inspections, asset management and inventory, tracking assets and field crews, conducting surveys, incident reporting, and managing parcel deliveries. This wide-ranging utility highlights Mobile GIS's important role in both enterprise and consumer applications.

Mobile GIS is a rapidly advancing field, evolving alongside edge computing, geographic information science (GIScience), and artificial intelligence (AI). It now incorporates frontier technologies, such as virtual reality (VR) and augmented reality (AR) on smartphones and wearable devices. VR offers users wearing specialized visual devices the ability to immerse themselves in 3D models, significantly enhancing their spatial understanding. AR enhances real-world views with GIS data overlays captured through a camera, enriching a user's perception of reality.

Additionally, Mobile GIS harnesses machine learning and geospatial artificial intelligence (GeoAI) to automatically detect and identify objects from photographs. Integration with large language models such as ChatGPT has led to the creation of ArcGIS Generative AI Assistants, which help compile layers, create web maps, and design smart forms specifically for mobile GIS applications. As GIS technology continues to integrate with advances in computing technologies and benefit from faster internet connections, its influence and applicability are expected to grow extensively.

Supporting technologies and design considerations

Mobile devices, mobile operating systems, and wireless communications form the foundation of Mobile GIS in the following ways.

- **Mobile devices:** This category includes smartphones, tablets, VR headsets, AR glasses, smart glasses, and smartwatches, which are essentially compact computers designed to be handheld, worn on the head, or worn on the wrist. The screen sizes of mobile devices are considerably smaller than those of desktop computers and can vary significantly.
- **Mobile operating systems:** The main operating systems include Android, iOS/iPadOS, and Linux. Specialty operating systems, such as Glass OS and Meta Horizon OS, are derivatives of Android, whereas Windows Holographic is based on Windows, and visionOS draws from iPadOS.
- **Wireless communication technologies:** Key technologies include Bluetooth, Wi-Fi, cellular networks, such as 5G, and satellite internet services, such as Starlink. Each technology varies in speed, range, and cost, serving different purposes (table 1.1). For clarity, the following discussion specifies typical formats and uses of these technologies.

Table 1.1. Comparison of speed, range, and common usage of wireless communication technologies

Wireless communication technologies	Speed	Range/coverage and common usage
Bluetooth	3 Mbps	10 meters. Facilitates communication between mobile devices and peripherals, such as GPS receivers and headsets.
Wi-Fi	Up to 9.6 Gbps (Wi-Fi 6)	100 meters, extendable. Used to build local area networks.
Cellular networks	Up to 20 Gbps (5G peak), 100 Mbps (5G average)	3–6.5 k per tower (5G). Coverage mainly in populated areas. Used for longer distance data connections, voice, and video calls.
Starlink	Between 25 and 220 Mbps, 100 Mbps average	Global coverage across most continents and oceans. Especially valuable in areas where Wi-Fi and cellular networks are unavailable.

The mobile platform offers significant opportunities for GIS, but it also presents unique challenges. The advantages of mobility come with constraints, including limited CPU speed, memory capacity, battery life, bandwidth and network connectivity, screen size, and keyboard dimensions. Although advancements in technology are mitigating some of these limitations, they still need to be carefully considered in the design and development of mobile GIS solutions.

- **Simplify user interface:** Design clear and intuitive interfaces. Avoid overcrowding the display with excessive toolbars and menus.
- **Streamline data entry:** Keep data forms simple and smart, displaying only relevant fields. Minimize text entry requirements by using drop-down lists and automated calculations wherever possible.
- **Optimize map visibility:** Use bold text labels, high contrast, and straightforward symbology to ensure maps are easily readable in outdoor settings.
- **Orient maps practically:** In navigation mode, align the map with the user's direction of movement rather than a fixed due-north orientation to enhance readability and relevance.
- **Implement offline support:** Ensure functionality in disconnected areas by preloading necessary data onto devices before entering these zones.
- **Choose appropriate development approaches:** For custom native apps, choose a specific platform or employ cross-platform programming languages to facilitate single-source code deployment across multiple platforms.

Mobile positioning technologies

The foundational element of Mobile GIS is location awareness, which has significantly evolved through various positioning technologies for outdoor and indoor environments. The importance of mobile location services was underscored by legislation such as the Enhanced 911 (E911) in the United States and similar laws globally. Before 2000, GPS technology was not commonly integrated into mobile phones, which relied primarily on cellular networks for locating callers. A tragic incident in Florida in 2001, in which a woman lost control of her car and ended up in a canal, illustrated the limitations of early mobile location technology. Unable to provide her location to the emergency operator, she tragically lost her life because the rescue units could not find her quickly. This incident highlighted the urgent need for precise mobile location services, prompting the introduction of laws mandating mobile providers to deliver the location of emergency calls with specific accuracy. these regulations have driven substantial advancements in mobile positioning technology.

Today's mobile positioning services use a combination of the following technologies (figure 1.1) to improve location accuracy, reliability, and availability.

- **Global Navigation Satellite System (GNSS):** Uses satellite signals to pinpoint precise geographic locations (longitude, latitude, altitude). Key systems include the US Global Positioning System (GPS), Russia's GLONASS, China's Beidou, and the European Union's Galileo. GNSS provides an average accuracy of 5 meters under clear skies, but this technology requires the line of sight between the GPS antenna and four or more satellites. Its accuracy can decrease because of satellite positions, cloud cover, and obstacles, such as buildings.
- **Cellular network-based positioning:** Relies on cellular network infrastructure to locate devices, using methods such as cell of origin and time difference of arrival. Although it offers broad coverage and is widely available, it is much less accurate compared with other positioning technologies.
- **Wi-Fi-based positioning:** Identifies device locations within about 100 meters of a Wi-Fi access point, with potential accuracy improvements to within 3 meters through Wi-Fi signal triangulation. This method depends on the availability of Wi-Fi signals and requires regular updates to Wi-Fi hot spot databases.
- **Bluetooth Low Energy (BLE):** Designed for low power consumption, BLE is used in tracking technologies such as Apple's AirTags, which are commonly used to attach to personal items for location sharing through apps. In buildings with dense BLE beacon setups, systems such as ArcGIS IPS™, an indoor positioning system, use algorithms to precisely locate mobile devices.

Figure 1.1. Mobile GIS uses positioning technologies to show your current location, often represented by a blue dot. This is achieved using a combination of the Global Navigation Satellite System (GNSS), cellular network-based Global System for Mobile Communications (GSM), Wi-Fi, Bluetooth Low Energy (BLE), Visual Light Communication (VLC), and other positioning technologies.

High-accuracy GPS receivers can reach submeter to centimeter location accuracy, depending on their ability to track and process satellite signals. GPS satellite signals are transmitted at different frequencies; the more frequencies a GPS receiver uses, and the more signals it receives, the more accurate it is. The accuracy can be further improved through real-time differential correction or postdifferential correction.

Mobile GIS architecture

Introducing Mobile GIS within the framework of Web GIS, which combines web technology with GIS, is crucial for appreciating its evolution and full capabilities. Initially in the mid-1990s, Mobile GIS mostly operated in a disconnected mode, in which operators loaded data onto devices each morning and, upon returning, docked the devices to transfer data back to a central computer.

However, with the rapid advancements in wireless communications, Mobile GIS is now primarily connected to the web, integrating seamlessly into Web GIS systems while still facilitating offline workflows. This connected approach allows Mobile GIS to continuously update servers with the latest field data. Conversely, web servers enhance Mobile GIS with comprehensive content and advanced analytics.

Esri® has two Web GIS products: ArcGIS Online and ArcGIS Enterprise, both offering similar functionalities (figure 1.2). ArcGIS Online is a cloud-hosted software-as-a-service

(SaaS) solution, entirely managed by Esri, which relieves organizations from the need to maintain hardware infrastructure. In contrast, ArcGIS Enterprise is a comprehensive Web GIS software suite that organizations can install and manage on various platforms, including Windows, Linux, and Kubernetes, whether on-premises or in cloud environments, such as AWS, Microsoft Azure, and Google Cloud.

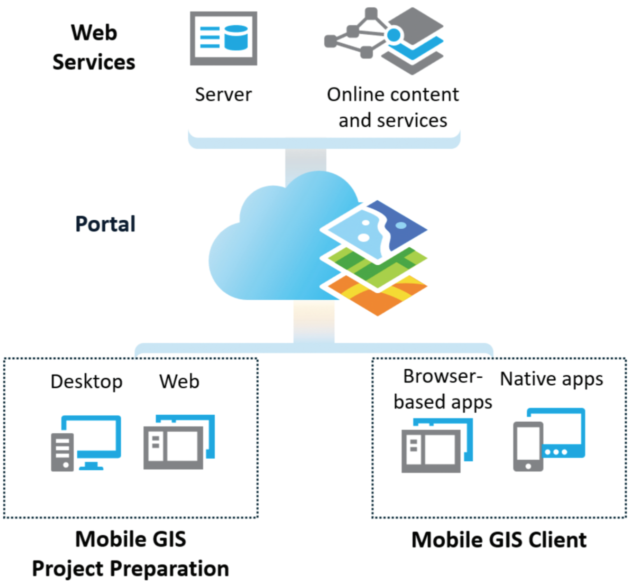

Figure 1.2. Mobile GIS is becoming an important component of Web GIS architecture. Web GIS provides back-end web services, content management, access control, and a series of apps for GIS professionals who create Mobile GIS solutions and for mobile workers who primarily use mobile apps.

Today's Mobile GIS is primarily designed as a component within Web GIS, featuring a cohesive architecture that enhances its functionality and accessibility in the following ways.

- **Portal centerpiece:** The core of the architecture is a portal—a part of either ArcGIS Online or ArcGIS Enterprise—which organizes, secures, and facilitates access to geographic information products, serving as a central hub for content management.
- **Back-end capabilities:** At the back end, GIS servers are deployed, allowing users to publish geospatial services. Additionally, there is access to ready-to-use content, such as ArcGIS Living Atlas of the World, which provides a repository of geospatial web services.
- **Client-side applications**
  - **GIS professionals:** Desktop and web clients are used by GIS professionals to design data schemas, publish services, and configure apps and projects for Mobile GIS.
  - **Mobile workers:** Mobile GIS clients are used by workers in the field, who can access layers and maps live when online or use preloaded content when operating offline.

ArcGIS suite of web and native apps for Mobile GIS

ArcGIS offers a comprehensive suite of web and native mobile apps designed for both field and indoor operations, enhancing the functionality and versatility of Mobile GIS.

Responsive web apps run inside web browsers and require data connection. This category includes ArcGIS Instant Apps, ArcGIS StoryMaps℠, ArcGIS Experience Builder, ArcGIS Dashboards, and ArcGIS Hub℠. these apps are responsive to mobile screens and allow configuration for optimal mobile layouts. Although not best suited for offline use, they work well in delivering field data for situational awareness and facilitating data collection and editing with reliable internet connection. Additionally, these browser-based apps are crucial for office-based tasks, such as data review, quality control, editing, approval, and assignment processes.

Native apps are downloaded and installed directly onto your device. ArcGIS offers a series of native apps, including the following.

- **Field Maps:** Integrates several earlier products, including ArcGIS Collector, ArcGIS Explorer, ArcGIS Tracker, and ArcGIS Workforce. Field Maps serves as an all-in-one solution that supports planning, recording and sharing location, understanding, and map-centric data capture.
- **ArcGIS Survey123:** Known for its simplicity and intuitiveness, this formcentric app allows users to create, share, and analyze surveys. It features smart forms with skip logic, defaults, flexible formulas, and powerful data-pulling capabilities. The app facilitates easy data collection through the web or mobile devices in any environment, requiring minimal training.
- **ArcGIS QuickCapture:** An ideal solution for rapid data collection features a user interface with large buttons, designed to minimize interaction with the device during fast-paced field data collection. It supports offline and online work, location sharing, mapping, and data collection.
- **ArcGIS Indoors™ mobile app:** Provides an indoor mapping experience to help users navigate their organization's indoor environment. Features such as wayfinding, routing, and location-sharing enhance connectivity to the workplace or campus and integrate with calendars for seamless navigation to meetings. Users can also report incidents related to indoor assets or locations.
- **ArcGIS Mission Responder:** Part of the ArcGIS Mission product, this app allows field users to participate in missions, offering tactical situational awareness and enabling communication and collaboration through real-time messaging and reporting.
- **ArcGIS Earth:** An interactive 3D tool for planning, visualizing, and evaluating events on the globe, it offers situational awareness on desktop and mobile devices and supports data in various formats, including 3D models, KML/KMZ files, and TXT files. ArcGIS Earth allows users to quickly manipulate 3D data and collaborate for enhanced decision-making.

Four pillars of Mobile GIS capabilities

Although data capture is the most used capability, ArcGIS Mobile GIS products offer a broader range of functionalities, organized into four main pillars (figure 1.3), as follows.

- **Field awareness:** Mobile maps significantly enhance situational awareness in the field. Mobile apps and maps can be used anytime and anywhere they are needed. Field Maps includes exploratory map tools and contextual pop-ups that deliver detailed information, including charts, documents, and photos, essential for on-site decision-making. Geofencing alerts mobile workers when they enter or leave designated areas. Integrated routing, navigation, and wayfinding capabilities efficiently locate assets, whereas location sharing allows workers to view, record, and share their locations. Field Maps also supports advanced capabilities, including viewing and tracing utility networks, measuring distances, and placing points by distance along linear referenced features.
- **Data capture:** The primary use of Mobile GIS. ArcGIS provides Field Maps, Survey123, and QuickCapture, each offering unique user experiences. these apps use both indoor and outdoor locations and can integrate with external, high-accuracy GNSS receivers for sub-foot accuracy. Smart forms eliminate paper-based workflows by improving the accuracy and speed of data collection through conditional logic, pick lists, and calculated expressions. Captured photos document and automatically integrate data of various formats into forms. This functionality is based on the layers and tables in Web GIS, with device edits syncing to the enterprise GIS for immediate availability.
- **Planning and coordination:** Field Maps and Survey123 enhance field operations by supporting mobile project planning, data preparation, task or to-do list assignments, prioritizing tasks, visualizing progress, and sharing the live location of field crews. they also enable communication between the field and the office by pushing notifications and messages.
- **Integration:** The full potential of mobile GIS capabilities is realized through integration with broader organizational systems. Data captured in the field feeds into enterprise information systems, enabling analysis and dissemination of information across the organization. Tools such as ArcGIS API for Python, ArcPy™, and webhooks can automate the integration, streamline tasks, generate reports, and optimize field workflows.